DAY 2 — PUBLIC IP vs PRIVATE IP ADDRESSING

INTRODUCTION

Every device connected to a network requires an IP address for communication.

🌐 What is an IP Address?

✅ Definition

An IP Address (Internet Protocol Address) is a unique numerical address assigned to a device in a network for communication.

It helps devices:

- Identify each other
- Send data
- Receive data
- Communicate over networks and the internet

▯ Simple Understanding

Just like:
📫 House address helps identify your home

Similarly:
🌐 IP address helps identify a device on a network.

💻 Example IP Address

IPv4 Example: 192.168.1.1

IPv6 Example: 2401:4900:97c2:aa1e::abcd

🎯 Main Purpose of IP Address

IP addresses are used for:
✅ Device identification
✅ Communication between devices
✅ Internet access
✅ Data routing
✅ Network management

🌍 Types of IP Addresses

Type	Purpose
Private IP	Used inside local networks

Type	Purpose
Public IP	Used on internet
IPv4	32-bit address
IPv6	128-bit address

IP addresses are categorized into:

1. Public IP Address
2. Private IP Address

Understanding these concepts is important in networking, cybersecurity, and Identity & Access Management (IAM).

What is a Public IP Address?

A Public IP Address is an IP address **visible** on the internet.

It is assigned by the **Internet Service Provider (ISP).**

Public IPs allow communication between devices over the internet.

Example:
106.192.36.233

Characteristics:

- Globally unique
- Internet-visible
- Assigned by ISP
- Used for internet communication

What is a Private IP Address?

A Private IP Address is used **inside** local/private networks.

Private IPs are not directly accessible on the internet.

Example:
10.221.64.26

Characteristics:

- Used internally
- Assigned by router
- More secure internally
- Reusable in different organizations

Private IP Ranges

10.0.0.0 – 10.255.255.255

172.16.0.0 – 172.31.255.255

192.168.0.0 – 192.168.255.255

Difference Between Public and Private IP

Public IP	Private IP
Visible on internet	Used inside local network
Assigned by ISP	Assigned by router
Globally unique	Local use only
Used for internet communication	Used for internal communication

What is NAT?

NAT stands for Network Address Translation.

NAT converts:
Private IP → Public IP

Example:
10.221.64.26
↓
106.192.36.233

NAT allows multiple devices inside a network to share one public IP address.

Why NAT is Important

NAT helps:

- Save IPv4 addresses
- Improve security
- Hide internal IP addresses
- Allow internet access for private devices

What is ISP?

ISP stands for Internet Service Provider.

Examples:

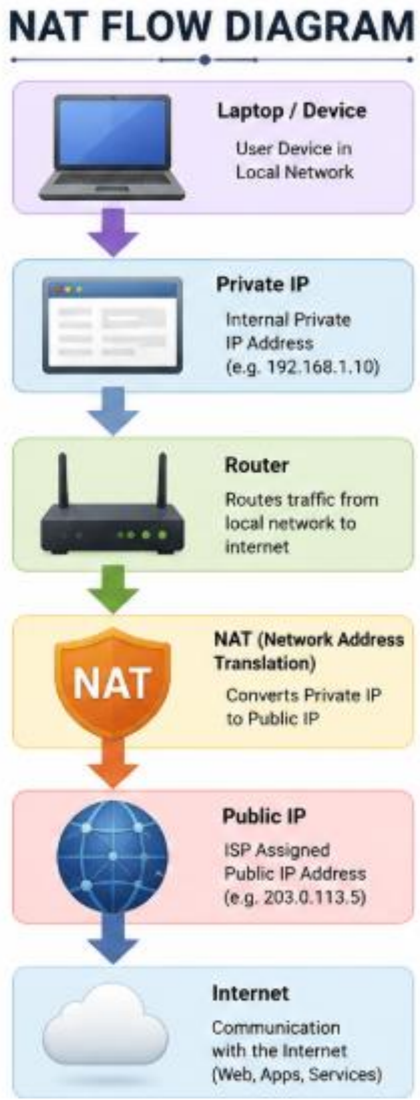
- Airtel
- Jio

- BSNL
- ACT

ISP assigns public IP addresses to users.

Internet Communication Flow

Laptop → Router/Gateway → Public IP → Internet



IAM Security Connection

IAM systems use IP addresses for:

- Login monitoring
- Geo-location tracking
- Conditional access
- VPN restrictions
- Suspicious login detection
- MFA triggering

Example:
If login occurs from another country, IAM may trigger MFA or block access.

Commands Practiced

ipconfig
ping google.com
nslookup google.com

Key Learning Outcome

Through this topic, I learned:

- Difference between public and private IP addresses
- NAT functionality
- ISP role in networking
- Internet communication flow
- Importance of IP addresses in IAM security